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Key Points:

- Resistivity and permeability of fractured rock are modeled as the fractures within the network are opened
- Fractured rock demonstrates percolation behavior similar to individual fractures
- Percolation behavior depends on density of fractures as well as how open they are

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Three-dimensional resistor network modeling of the resistivity and permeability of fractured rocks

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Abstract The resistivity and permeability of 3-D fracture networks have been modeled as the fractures within these networks are incrementally opened. In these models, the ratio of the rock resistivity to that of the fluid is 10^4 , and the rock permeability is $1 \times 10^{-18} \text{ m}^2$. The changes in both resistivity and permeability depend on characteristics of the network itself such as fracture density, as well as the aperture distribution within individual fractures. In dense fracture networks where the density constant α is equal to 30, a percolation threshold can be defined in terms of mean fracture aperture, below which both resistivity and permeability are close to their matrix values. At the percolation threshold, a change in mean aperture of 0.02 mm can change the permeability by 4 orders of magnitude and resistivity by a factor of 4. The percolation threshold does not necessarily occur at the same aperture for different flow directions, so fracture networks near their percolation threshold commonly show anisotropy in both resistivity and permeability. Most sparse networks (α equal to 0.3 or less) do not percolate no matter how open the fractures are. On the other hand, many fracture networks with an intermediate value of α (equal to around 3) will percolate only in one or two directions. This can lead to very strong anisotropy in both resistivity and permeability (up to a factor of 160 and 10^9 , respectively), with anisotropy increasing as the aperture of the fractures increases.

Plain Language Summary Knowledge of the distribution of the subsurface permeability, or the ability of rocks to allow fluid to pass through them, is important in managing resources such as geothermal, oil and gas, and groundwater. However, permeability varies strongly from place to place and cannot always be successfully mapped by standard industry tools such as seismic reflection. Methods that measure the electrical resistivity of rocks may be useful in determining the permeability if there is a known relationship between these two properties. We use synthetic models to better understand the relationship between resistivity and permeability in fractured rocks. We find that both resistivity and permeability are strongly controlled by both the density of fractures in a rock and how open these fractures are. There is a critical aperture, above which both permeability and resistivity increase strongly, depending on how well connected the fault networks are.

1. Introduction

Fracture permeability is vital to the successful development of many unconventional energy resources, such as geothermal and unconventional hydrocarbons [e.g., Barton *et al.*, 1997; Barelli *et al.*, 2000; Cumming, 2009; Muñoz, 2014]. These resources are often located at several kilometers depth, where compaction and cementation processes can remove much of the primary permeability [e.g., Barton *et al.*, 1997; Cumming, 2009]. Exploration of fracture permeability, however, is not straightforward. Permeability can vary by orders of magnitude over length scales of kilometers to meters, making it difficult to target enhanced permeability zones from the surface.

Electrical resistivity has long been used to infer the porosity of rocks, for example, in Archie's law, a widely used empirical relationship that relates resistivity to porosity and water saturation in a rock with high background resistivity [Archie, 1942]. Variants of Archie's law exist, including to account for nonnegligible surface conduction [e.g., Tiab and Donaldson, 2012] or for multiple fluid phases [e.g., Glover, 2010]. The porosities can then be empirically related to permeability. However, Archie's law was developed for porous rocks with high primary permeability and may not be relevant for low porosity, fractured rocks.

The magnetotelluric method is sensitive to changes in the subsurface resistivity both spatially and with orientation. It has been shown to have merit in detecting resistivity changes due to both addition of electrically conductive fluids in an enhanced geothermal system [e.g., Peacock *et al.*, 2012, 2013; Rosas-Carbajal *et al.*, 2015] and more recently, hydraulic stimulation of a shale gas prospect [e.g., Rees *et al.*, 2016a, 2016b]. In both examples, the addition of fluids was associated with increases in permeability along the fracture network. However, quantitatively linking the resistivities to permeability has not yet been possible.

Kirkby *et al.* [2016] modeled the resistivity and permeability of individual synthetic fractures, allowing a link to be made between the two properties. The fractures were generated using the fractal model of Brown [1989]. This approach assumed fracture surfaces with isotropic roughness. The roughness was controlled by fractal dimension, which was assumed to be constant, with a value of 2.5 based on measured characteristics in rock surfaces. The modeling was carried out at various points as the fractures were incrementally separated, providing information on the evolution of both resistivity and permeability as a fracture is opened.

Both the permeability and resistivity of the fractures were shown to demonstrate percolation behavior. At low apertures, fluids are only present in isolated pockets, and both the permeability and resistivity are close to the rock matrix values. However, as the fracture is opened, the void space becomes more connected, eventually reaching its percolation threshold, when a fully connected pathway through the fracture develops. At this point, both the permeability and resistivity increase rapidly. The modeling showed that in order to have a substantial reduction in resistivity (by greater than a factor of 2) and permeability enhancement (greater than 1 order of magnitude) from fractures, it is necessary that the fracture is at or over its percolation threshold, confirming results from previous studies [e.g., Bahr, 1997]. The modeling also demonstrated the percolation threshold in terms of measurable fracture parameters such as the separation between the fracture planes. For the fracture surface model used by Kirkby *et al.* [2016], the change from below to above the percolation threshold occurs over an aperture change of <0.1 mm. Thus, very small changes in the fracture aperture can lead to large changes in both resistivity and permeability of a fracture.

One limitation of the work of Kirkby *et al.* [2016] is that it was carried out on individual fractures and not fracture networks. Thus, it is difficult to directly relate the results to resistivities and permeabilities that might be observed in the subsurface. There are empirical relationships that relate the spacing, size, and offset of fractures within networks [e.g., Bonnet *et al.*, 2001; Kim and Sanderson, 2005]. These could be used to build on the work of Kirkby *et al.* [2016] to gain a more complete understanding on the effect fracture networks have on the resistivity and permeability values that are measured in the subsurface.

In this paper, we build on the work of Kirkby *et al.* [2016] by modeling the electrical and fluid flow properties of synthetic fracture networks in rocks. Fault networks are constructed based on established relationships between fracture spacing, size, and offset. Each fracture within the network is constructed using the same roughness model as that used by Kirkby *et al.* [2016], with the parameters chosen based on measurements on real rock fractures. As with Kirkby *et al.* [2016], the resistivity and permeability of each of these fracture networks are evaluated at several points as the aperture of the fractures within the network is incrementally increased.

2. Background

2.1. Resistor Networks

The approach used in this work is based on the resistor network approach used by Bahr [1997]. Electrical current and fluid flow are assumed to occur through a network of resistors. In the approach of Bahr [1997], the resistors were assumed to either be perfect conductors or insulators. Kirkby *et al.* [2016] developed this concept further by taking the resistance of each resistor to be related to the local fracture aperture and, in the case of resistivity, the conductivity of the fill material. A potential difference is then simulated and used to determine the electrical current passing through the network. This current distribution is then used in conjunction with the potential difference to calculate the bulk resistivity. Likewise, a pressure difference is applied across the network to determine the fluid flow distribution, which is then used to calculate the permeability.

The resistor network approach is a simplification of the hydraulic and electrical processes occurring in porous media. It assumes laminar fluid flow, which is a reasonable assumption for most natural groundwater flow rates [Reynolds, 1883; Garven, 1995; Drazin and Reid, 2004; Kirkby *et al.*, 2016], but may not be valid at higher flow rates, such as those encountered during hydraulic stimulation.

In addition, electrical conductivity is a tensor property with both in-phase and quadrature (out of phase) components [Vinegar and Waxman, 1984]. It is also frequency dependent, although the frequency dependence is normally weak for the in-phase components [Revil, 2012; Revil et al., 2015]. The resistor network approach simplifies these aspects as it tells us about the in-phase component but nothing about the quadrature components and provides only a single estimate of resistivity with no frequency dependence. Furthermore, the electrical conduction in a pore fluid-rock mixture has a surface component, resulting from movement of ions along the pore water-rock interface [Waxman et al., 1968; Bussian, 1983]. The surface component is greatest when clay minerals are present and is less important in the absence of clay [Waxman et al., 1968]. The amount of surface conduction depends on the composition of the fluid and clay and is also temperature dependent [e.g., Waxman et al., 1968; Revil et al., 1998; Revil, 2012]. We have not included surface conductivity in the resistor network models presented here. Thus, the models presented here are most relevant to rocks containing little or no clay and high fluid salinities.

2.2. Hydraulic and Electric Resistance in Fractures

The electrical resistance at each point in the fracture, R_E , is related to the local current I and the electrical potential v via Ohm's law [Brown, 1989; Kirkby et al., 2016] as follows:

$$I = \frac{1}{R_E} \nabla v \quad (1)$$

where

$$R_E = \frac{\rho_f \Delta z}{A} = \frac{\rho_f \Delta z}{b \Delta y} \quad (2)$$

where ρ_f is the fluid resistivity, A is the cross-sectional area perpendicular to flow, b is the fracture aperture, Δz is the cell size in the direction of flow, and Δy is the cell size perpendicular to both the flow direction and the fracture opening direction [Brown, 1989; Kirkby et al., 2016].

Likewise, the permeability k is related to the local fluid flow rate Q and the pressure p through Darcy's law

$$Q = \frac{kA}{\mu} \nabla p \quad (3)$$

where μ is the fluid viscosity. Fluid flow in fractures is commonly approximated by the local parallel plate model, in which the fracture surfaces are assumed, locally, to be flat and parallel. The steady state solution of the Navier-Stokes equations for laminar fluid flow leads to a cubic dependence of fluid flow on aperture [e.g., Witherspoon et al., 1980; Brown, 1989]

$$Q = -\Delta y \frac{b^3}{12\mu} \nabla p \quad (4)$$

From equations (3) and (4) we obtain the permeability as $b^2/12$ and the local hydraulic resistance, R_H , as

$$R_H = \frac{12\mu \Delta z}{\Delta y b^3} \quad (5)$$

The local parallel plate model for fluid flow can be extended to nonparallel, sloping plates through a correction factor derived by Brush and Thomson [2003]. The correction is based on the Navier-Stokes solution for fluid flow in a wedge and also corrects for extra path length resulting from sloping plates. It was shown to provide a good approximation to the full Navier-Stokes solutions at the low flow velocities likely to be encountered in natural groundwater flow [Brush and Thomson, 2003]. This correction was used by Kirkby et al. [2016] who developed and applied an equivalent correction for electrical aperture.

2.3. Scaling in Fracture Networks

Bonnet et al. [2001] reviewed relationships in the scaling between fracture length, spacing, displacement, and aperture in natural fracture networks. The authors concluded that power law distributions are the most widely applicable models for characterizing the size distribution of most fracture systems, although in some cases (for example, where there is horizontal bedding), exponential or lognormal distributions may apply [Odling et al., 1999].

Assuming a power law distribution, the total number of fractures per area R^2 , $N_{f,2-D}$ greater than length l_{min} , can be modeled as follows:

$$N_{f,2-D} = \int_{l_{min}}^{l_{max}} \alpha l^{-a} R^2 dl \quad (6)$$

[Bonnet *et al.*, 2001]. The constant α and exponent a are determined empirically. The truncation length, l_{min} , is usually assumed to be the length of the smallest observed fracture. In practice, this length depends on the resolution, which depends on the method and scale of observation. Bonnet *et al.* [2001] presented a compilation of the characteristics of natural fracture networks from 45 separate studies. The fractures range in length from the core scale (~ 3 mm to 1 m scale) to the aerial photo/seismic scale (~ 10 m to 100 km).

The number of fractures in a particular length range within a given area is then given by

$$N_{f,2-D} = \frac{\alpha}{a-1} (l_{min}^{1-a} - l_{max}^{1-a}) R^2 \quad (7)$$

Bonnet *et al.* [2001] presented α and a values calculated from the 45 natural fracture networks included in the review. All of the presented fracture networks were calculated from 2-D areas, i.e., in cross section or from maps.

There is a moderately high degree of scatter in observed values of a . The exponent a ranges from 1.3 to 3.3 with most values clustering within the 1.75–3.0 range. However, of the 45 studies in the compilation, only six were calculated for fracture lengths that extend down to 1 m or less [Reches, 1986; Bahat, 1987; Knott *et al.*, 1996; Schlische *et al.*, 1996; Ackermann and Schlische, 1997], and in all but one of these, a falls within the range 2.0–2.75.

The constant, α , shows an even higher degree of scatter, ranging from 10^{-5} to 100, although it broadly decreases as the fracture network area increases. Of the six presented studies that include fracture lengths < 1 m, α ranges from around 0.3 to just under 100, with a median of approximately 3.

Both a and α were calculated from 2-D fracture distributions [Bonnet *et al.*, 2001]. Given that the fracture networks in this study are 3-D, equivalent values for 3-D need to be calculated. Synthetic tests were carried out by generating 3-D fracture networks and then analyzing the fracture length distribution of 2-D slices intersecting them. These tests showed that the equivalent value of a for 3-D can be obtained by adding 1 to a (2-D). The value of α is the same in 3-D as it is in 2-D; however, the area R^2 becomes a volume (R^3). As such, the number of fractures in a volume R^3 , $N_{f,3-D}$ is given by

$$N_{f,3-D} = \frac{\alpha}{a} (l_{min}^{-a} R^3) \quad (8)$$

2.4. Fracture Aperture

Both the hydraulic and electrical resistance in each fracture plane in the fracture network depend on the aperture distribution within that fracture plane (section 2.2). Thus, it is important to accurately model the aperture distribution in order to adequately represent these two properties.

Fracture surfaces have power spectral densities of the form

$$G(f) \propto f^{-\beta} \quad (9)$$

where G is the power spectral density, f is the spatial frequency, and β is related to the fractal dimension D through [Brown, 1987]

$$\beta = \frac{7-D}{2} \quad (10)$$

Fractal dimension ranges from about 2.2 to 2.6 in rock fractures [e.g., Brown, 1995; Glover *et al.*, 1998a; Matsuki *et al.*, 2006; Ishibashi *et al.*, 2015]. Brown [1995] used this relationship to develop an algorithm for generating self-similar fracture surfaces, which uses the inverse Fourier transform to generate fractures with power spectral density functions described by equations (9) and (10). The phase of each Fourier component was scaled by a random number between zero and one, in order to replicate the random phase spectra that has been observed in natural rock fractures [Brown, 1995].

The random phase can be used to develop mismatching between two opposing fracture surfaces on different length scales. If the random phase components are the same between the two fracture surfaces, then the two

fracture surfaces will be identical. If they are different for some frequencies, then the surfaces will show mismatching over the corresponding length scales. Partial correlation can be achieved by combining the random phase component from one surface with a new random component for the second surface using a weighted mean. The weighting factor γ is normally defined as some function of frequency [Glover *et al.*, 1998a; Matsuki *et al.*, 2006]. A mismatch cutoff frequency can be defined, which describes the length scales over which the mismatching develops, and has been shown to range from about 0.25 to 8.0 mm in real rock fractures [Glover *et al.*, 1997, 1998a, 1998b; Matsuki *et al.*, 2006; Ogilvie *et al.*, 2006; Ishibashi *et al.*, 2015]. Various models have been applied to describe the shape of the cutoff, and these are discussed by Kirkby *et al.* [2016]. We follow Kirkby *et al.* [2016] by applying a mismatch cutoff frequency of 1 mm. Below the frequency cutoff, γ is a linear function of frequency (section 3.2).

The absolute amplitude of surface height variations, measured in terms of the standard deviation σ_h , scales with the length L of a profile across the fracture [Matsuki *et al.*, 2006]

$$\sigma_h = \sigma_{h,\text{ref}} \left(\frac{L}{L_{\text{ref}}} \right)^{3-D} = s L^{3-D} \quad (11)$$

where L_{ref} and $\sigma_{h,\text{ref}}$ are the length and standard deviation of the heights for a reference fracture surface. Kirkby *et al.* [2016] combined these to give s , a fracture length-independent scaling factor

$$s = \frac{\sigma_{h,\text{ref}}}{L_{\text{ref}}^{3-D}} \quad (12)$$

Kirkby *et al.* [2016] demonstrated that there is broadly an inverse relationship between the two parameters. That is, in fracture surfaces with high fractal dimensions the standard deviation of the surface height overall tends to be smaller. Kirkby *et al.* [2016] used a fractal dimension of 2.4, which corresponds to a scaling factor of 1.9×10^{-3} .

3. Method

3.1. Fault Network Generation

A small volume of 1.5 cm $\times$ 1.5 cm $\times$ 1.5 cm was used. As discussed later, this volume size is a compromise between being able to model multiple realizations of a fracture network in a reasonable time frame (given the number of cells required) and modeling a volume of rock that is large enough to be relevant to measurements of resistivity and permeability in the subsurface.

The fracture network is composed of three orthogonal sets of fractures. While this is not necessarily realistic for many natural fracture networks, it is a reasonable starting point in beginning to understand the cumulative effect fractures of different sizes have on a rock volume.

To generate synthetic fracture networks, an approach similar to that used by Xu and Dowd [2010] was used. The general approach is that the position of each fracture in a network is defined randomly and assigned attributes including size, orientation, and roughness. If details of the fracture network are known (e.g., the dominant fracture orientation or fracture locations) then these can be assigned. However, for this work, the fracture networks being described are completely synthetic, so key characteristics are selected from observations of natural fracture networks.

First, equation (8) was used to determine a fracture size distribution. This was discretized using 20 bins per decade in log space, with equation (8) providing the number of fractures in each length bin. Then, for each fracture, three uniform random seeds were generated to define the x , y , and z positions of the centre point of the fracture within the rock volume. A uniform random fracture length was selected from within the length bounds defined by the bin, and this length was also used as the height. An orientation was also randomly selected with equal probability for the yz , xz , and xy planes. The fracture was then assigned to the rock volume. Figure 1 shows the resulting distribution of fractures for one of these fracture networks.

3.2. Aperture Assignment

Each fracture in the network was assigned an aperture using the method described by Kirkby *et al.* [2016], which follows that of Matsuki *et al.* [2006] and Brown [1995]. Two partly correlated fractal fracture surfaces were generated for each fracture, and these were then separated by a fixed distance for modeling.

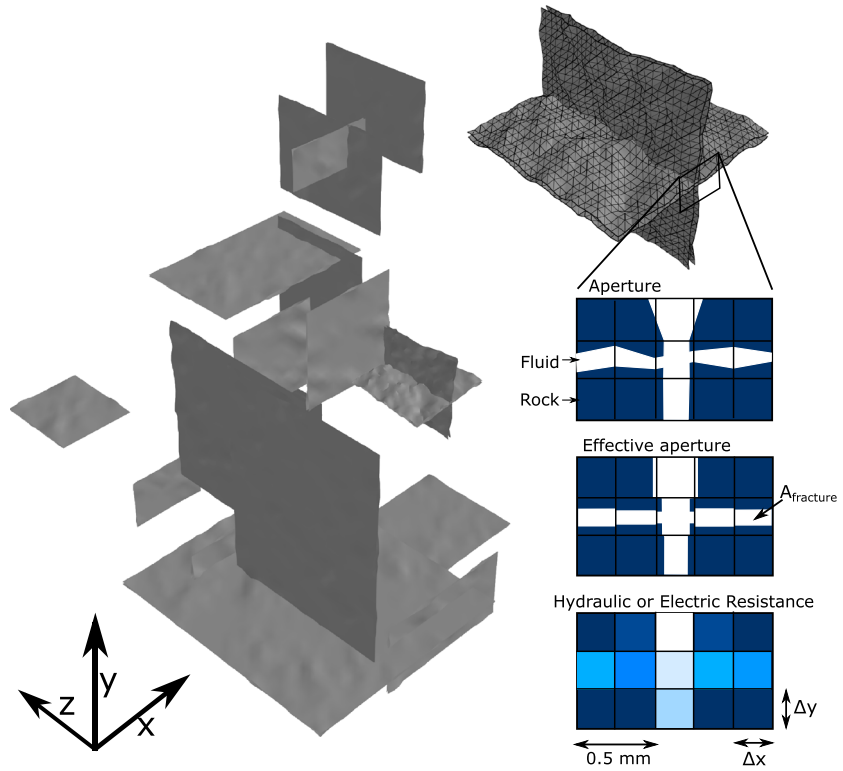


Figure 1. Key features of the model setup used for modeling fluid flow and current through fracture networks. An example fracture network is shown, generated with a equal to 2.5 and α equal to 3. Two intersecting fractures from the network are shown in detail, from the fracture surfaces for each fracture to the aperture, corrected aperture, and finally, the resistance. All aperture values in this diagram represent z direction resistances, i.e., into the page. The cell size in the x and y directions is indicated, along with the local area of the fracture, A_{fracture} , indicated by the white area in each cell. The vertical fracture on the right-hand side has a local aperture that exceeds one cell width, and so the fracture has been extended into the adjacent cells.

The Fourier component a of each fracture surface is related to the fractal dimension D and the spatial frequencies p and q in the x and y directions, where l is the size of the fracture through

$$a_{1p,q} \propto (p^2 + q^2)^{((4-D)/2)} e^{j2\pi R_1} \quad (13)$$

$$a_{2p,q} \propto (p^2 + q^2)^{((4-D)/2)} e^{j2\pi(R_1 + \gamma R_2)} \quad (14)$$

where R_1 and R_2 are independent uniform random numbers between 0 and 1 [Matsuki *et al.*, 2006].

The surfaces differ by the random component R_2 , scaled by γ , which we define based on the mismatch cutoff frequency f_c (section 2.4)

Once the two fracture surfaces had been created, they were scaled according to equation (11) and separated by a fixed value for modeling. Overlapping parts of the fracture resulting in negative apertures were set to zero. Geometry-corrected apertures were then calculated using the method of Kirkby *et al.* [2016].

3.3. Local Resistance

The apertures were then used to calculate local electric and hydraulic resistances associated with the fracture plane, using equations (2) and (5). To calculate the local resistance of the cell, a weighted harmonic mean of the resistance associated with the fracture, and that associated with the surrounding matrix in that cell, was used.

The fact that the modeling is now being carried out in 3-D presents several additional challenges in assignment of aperture and calculation of local resistances. In 3-D, it is not possible to dynamically change the cell size perpendicular to each fracture according to the fracture aperture, as was done by Kirkby *et al.* [2016], as this would change the frequency spectrum of fracture surfaces in the other directions. To counteract this, local fracture apertures wider than one cell width were expanded across to fill adjacent cells. The expansion

was done symmetrically, i.e., the expanded fractures took up three, five, or seven cells and not two, four, or six (Figure 1). In the case of electrical resistance, the fracture volume simply expands to adjacent cells, resulting in new apertures that can be used directly in equation (2). The hydraulic resistance calculation is more complicated. This is because the resistance is related to the cube of the aperture, and so if the local aperture in each cell was reduced to fit in that cell, this would reduce the resistance of that cell by the cube of the aperture.

To manage hydraulic resistances for cells with the local aperture greater than the cell size, a concept of hydraulic resistivity is introduced. The local hydraulic resistivity, ρ_h , in any cell can be derived from equations (2) and (5). For a connector in the z direction, associated with a fracture opening in the x direction, ρ_h is given by

$$\rho_h = \frac{R_H A}{\Delta z} = \frac{12\mu\Delta y}{b^2\Delta z} \quad (15)$$

where Δy and Δz are the cell sizes in the y and z directions, respectively. If the cell size is the same in all directions, then ρ_h is simply $12\mu/b^2 = k\mu$, where k is the permeability. Thus, the hydraulic resistivity for matrix cells is simply $k_m\mu$, and in cells that contain both matrix and fracture, a weighted harmonic mean is used as for electrical resistivity. The hydraulic resistivity can then be used to calculate the resistance of the cell, in a similar way to electrical resistivity.

The second challenge is handling fracture intersections. To assist with calculation, each connector (in each of the x , y , and z directions) was initially assigned three aperture values representing fractures opening in the x , y , and z directions. The local resistance value was calculated for each cell, in each direction, by combining the resistances associated with each fracture opening direction (Figure 1). In the case of electrical resistance, this was simply achieved by using the area of each of the fractured and matrix parts of the connector, and the resistivity of the fluid and matrix, in equation (2). For example, for a connector in the z direction, with fluid resistivity ρ_f and rock matrix resistivity ρ_m

$$R_{E,z} = \left(\frac{A_{\text{fracture}}}{\rho_f \Delta z} + \frac{\Delta x \Delta y - A_{\text{fracture}}}{\rho_m \Delta z} \right)^{-1} \quad (16)$$

where the area taken up by the fracture, A_{fracture} , is given by

$$A_{\text{fracture}} = b_x \times \Delta y + b_y \times (\Delta x - b_x) \quad (17)$$

where b_x and b_y are apertures opening in the x and y directions. In the case of hydraulic resistance, this was again more complicated. The hydraulic resistance in cells where two fractures intersect was defined by summing the local hydraulic resistance associated with each of the two intersecting fractures and then subtracting the intersecting part. Equation (15) implies that the local hydraulic resistivity for each of the two intersecting fractures will not necessarily be the same. The hydraulic resistivity associated with the smaller aperture (higher hydraulic resistivity) was therefore subtracted. For example, for a connector in the z direction

$$R_{H,z} = \left(\frac{b_x \Delta y}{\rho_{h,zx}} + \frac{b_y \Delta x}{\rho_{h,zy}} - \frac{b_x b_y}{\max(\rho_{h,zx}, \rho_{h,zy})} + \frac{\Delta x \Delta y - b_x \Delta y - b_y \Delta x}{k_m \mu} \right) \quad (18)$$

where $\rho_{h,zx}$ and $\rho_{h,zy}$ are hydraulic resistivity values in the z direction, associated with fractures opening in the x and y directions, respectively.

3.4. Input Parameters

A cell size of 0.25 mm was used for the networks, as was used by Kirkby *et al.* [2016]. Using this cell size, it was possible to adequately represent the mismatching behavior of the fractures, which occurs over wavelengths > 1 mm (section 2.4), while at greater cell sizes this was not possible.

The largest resistor network that could be computed in a practical amount of time was around $60 \times 60 \times 60$. This size can be calculated in around 20 min, while increasing the number of cells to $80 \times 80 \times 80$ increases computation time to around 9 h. Given that several thousands of models were to be run over the different input parameters as detailed below, a $60 \times 60 \times 60$ network (or $1.5 \text{ cm} \times 1.5 \text{ cm} \times 1.5 \text{ cm}$) was considered to be a reasonable compromise.

The parameters controlling the distribution of fractures in the network are maximum and minimum fracture length, the exponent in equation (8), a , and the constant α . A minimum fracture length (truncation length) of

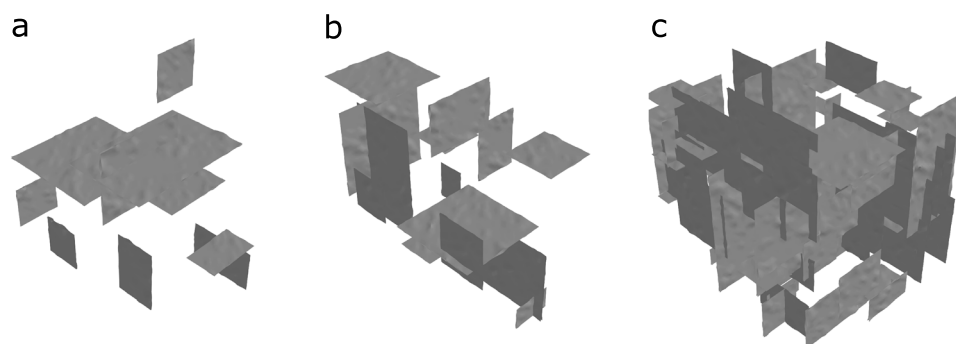


Figure 2. Examples of the fracture distribution for networks generated using three values of the fracture network density constant α , (a) α equal to 0.3, (b) α equal to 3, and (c) α equal to 30.

3 mm was used, based on the minimum fracture length observed in natural fracture networks in the review of Bonnet *et al.* [2001]. The maximum fracture length was taken as the size of the fracture network.

Three values of α were used: 0.3, 3, and 30. These were selected to reflect the distribution of α in Bonnet *et al.* [2001]. For this work, a was kept constant at 2.5. It is acknowledged that varying a will likely also have an effect on the permeability and resistivity of the fracture networks. However, given the higher degree of variability in α , it is anticipated that variation in α will have a greater effect. Future studies could potentially analyze how the parameter a affects the resistivity and permeability of fractured rocks. Figure 2 shows examples of fracture networks generated using alpha values of 0.3, 3, and 30.

Likewise, the parameters of the fracture planes were kept constant. As with Kirkby *et al.* [2016], a fractal dimension of 2.4, and a scaling factor of 1.9×10^{-3} , was used. These parameters, too, may affect the resistivity and permeability of the fracture networks; however, to keep the modeling as simple as possible, these were also kept constant.

The matrix permeability was set to $1 \times 10^{-18} \text{ m}^2$. Kirkby *et al.* [2016] showed that the main effect increasing the matrix permeability has is to raise the starting permeability below the percolation threshold. A rock to fluid resistivity ratio, defined here as m , of 10,000, was used. Future work will consider other m values; however, a value of 10,000 represents a starting point in analyzing the electrical and hydraulic properties of fractured rocks.

Each permutation was modeled over 20 different fracture separation values ranging from -0.16 mm to 0.89 mm . Fault separation was introduced by adding or subtracting a fixed value from the second surface generated in section 3.2. The modeling was then repeated 60 times using different random number seeds to generate both the distribution of fractures within the network and the fracture surfaces for each of the individual fractures. Given that each model was run six times, for the x , y , and z permeability and resistivity, this equates to a total of 6000 model runs.

4. Results

4.1. Densely Populated Fracture Networks

4.1.1. Resistivity and Permeability

Figure 3 shows the results of fracture network modeling over 20 different fracture separation values for sixty random fracture networks. Both resistivity and permeability are shown as a function of fracture separation for the highest value of α considered in this study, 30.

For α equal to 30 (Figure 3), all but one of the models demonstrate percolation behavior in all three directions. That is, as the fracture separation increases, the resistivity and permeability initially increase moderately; however, at some point a fully connected pathway develops and the permeability and resistivity increase more rapidly. The permeability-aperture curves show a similar pattern to that of individual fractures [Kirkby *et al.*, 2016], although the percolation threshold occurs at a slightly lower mean fracture aperture on average (just under 0.02 mm rather than $0.02\text{--}0.03 \text{ mm}$) and begins to level out at a slightly lower permeability. The resistivity percolation threshold also occurs just under 0.02 mm mean aperture. This curve is also similar to that for individual fractures but levels off later.

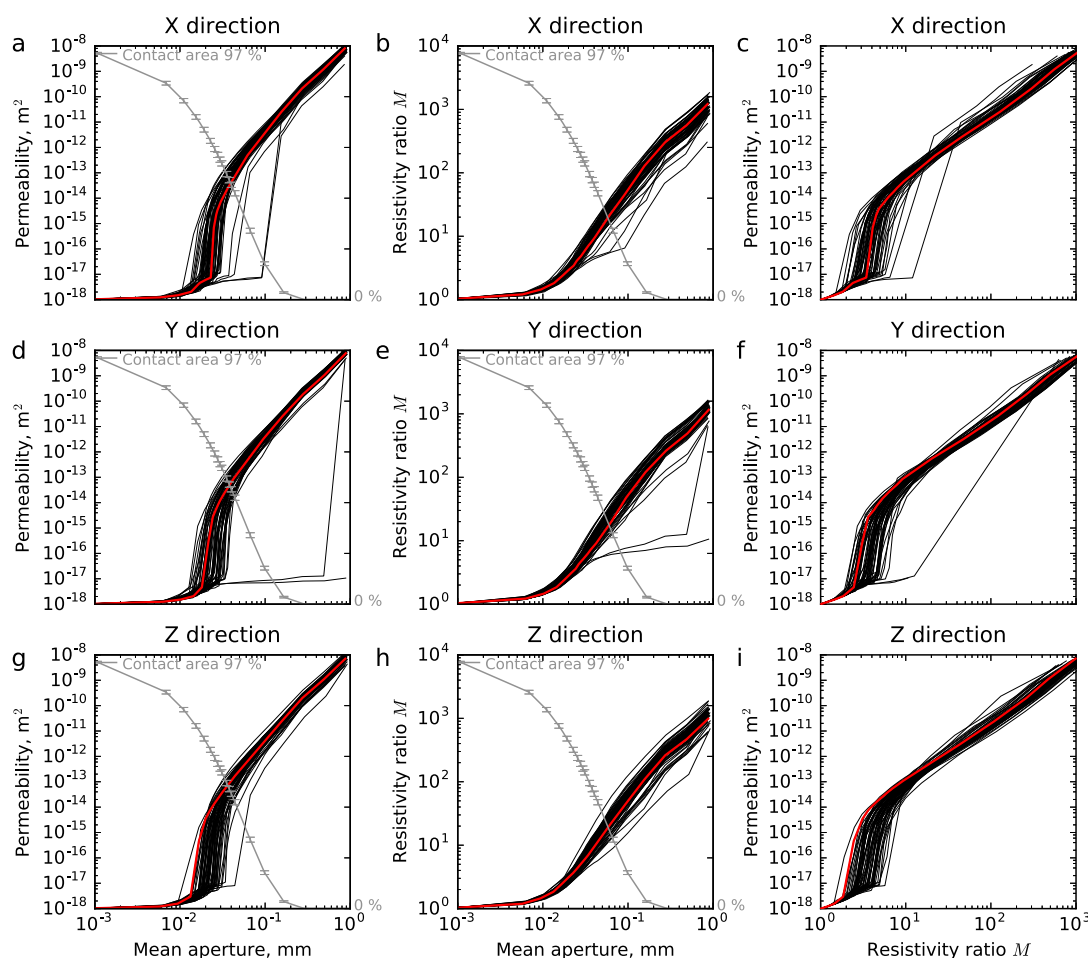


Figure 3. The evolution of permeability and resistivity in the x , y , and z directions (black lines) in a dense fracture network with the density constant α equal to 30, as the fractures within that network are opened. (a, d, and g) Permeability and (b, e, and h) resistivity ratio M , in three directions as a function of arithmetic mean aperture. The mean and standard deviation contact area for all models is shown in grey. (c, f, and i) Permeability as a function of M . The resistivity and permeability of the fracture network in Figure 5 are shown in red.

Permeability is also shown as a function of the rock matrix to fracture network resistivity ratio, which we define here as M (Figure 3c). As with individual fractures, there is a percolation threshold in permeability with respect to M . The main difference between the curve for an individual fracture and that for a fracture network is the location of the percolation threshold. In fracture networks with α equal to 30, the percolation threshold occurs at an M value between 2 and 10. So (given an m value of 10^4) if M is greater than about 2–10, the rock is likely to be above its percolation threshold for fluid flow.

The percolation threshold does not necessarily occur at the same fracture separation in each of the x , y , and z directions. As a result, near the percolation threshold, the resistivity and permeability in the individual fracture networks become anisotropic (Figure 4). A similar effect was noted by Bahr [1997] in modeling the connectivity 2-D resistor networks. However, this effect may simply reflect the limited size of the fracture networks considered in this study and may disappear at larger sizes.

4.1.2. Fluid and Current Flow

An example of the fluid and current distribution in one of the fracture networks with an α value of 30 is shown in Figure 5. The modeled current and fluid flow are shown for four different values of fracture separation. These diagrams demonstrate the physical processes that are occurring as the fractures within the network open and the network becomes more connected. The processes are quite similar to those described by Kirkby *et al.* [2016] for individual fracture planes.

At a fracture separation of -0.16 mm (i.e., the fracture planes in each fracture were shifted together by 0.16 mm, resulting in a mean aperture of 8×10^{-4} mm), there are no fully connected fractures, and as such,

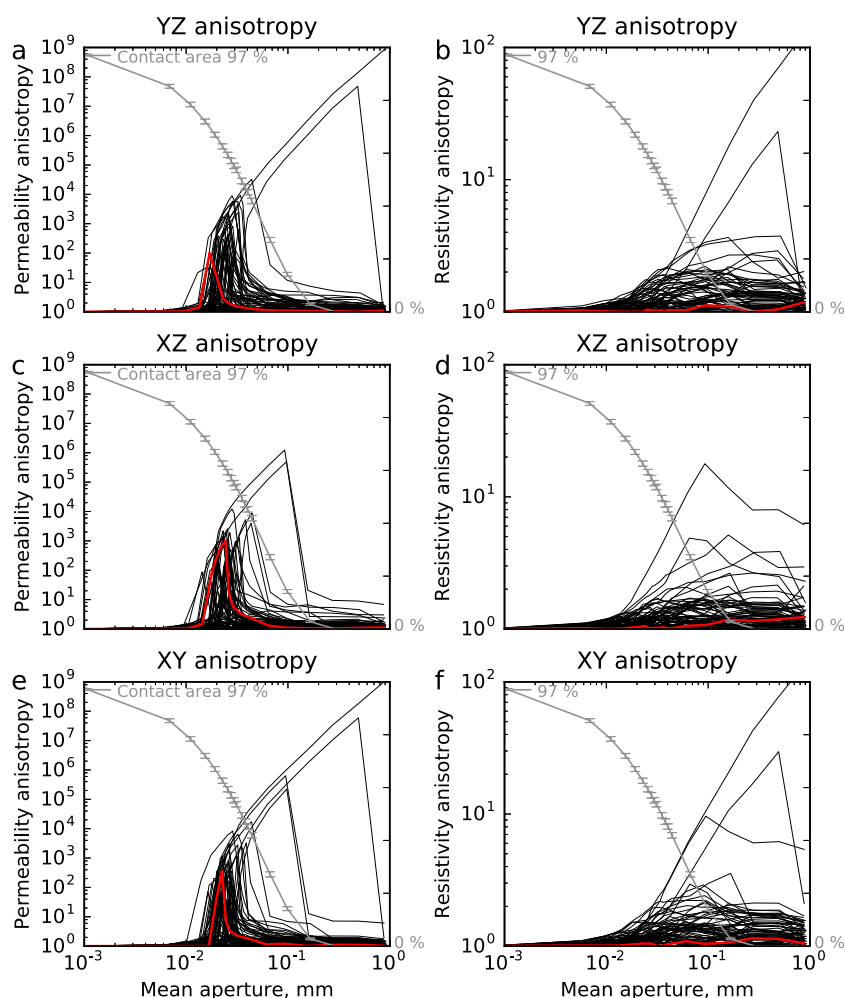


Figure 4. Anisotropy in permeability and resistivity as a function of arithmetic mean aperture for an α value of 30. Anisotropy shown as the maximum factor, i.e., xy anisotropy is the maximum of x/y and y/x resistivity or permeability. (a, c, and e) Permeability anisotropy. (b, d, and f) Resistivity anisotropy.

most of the conduction and fluid flow occur through the rock matrix. At this point, the contact area on the fractures is 97%, and the fracture porosity is 0.04%. The permeability and M have both increased by a small, identical amount, at $1.02 \times 10^{-18} \text{ m}^2$ and 1.02, respectively.

Once the fracture separation increases to -0.038 mm (mean aperture of 0.017 mm), connected pathways begin to form within the fracture network, although as yet, there are no fully connected pathways. Notably, Figure 5 shows both fluid flow and conduction in the matrix decreasing at this point, as fluids and current are no longer forced to pass through the rock matrix. These pathways travel along different intersecting fracture planes, changing direction as they reach different fracture planes. The mean fracture contact area is now 68% and the fracture porosity 0.9%. The permeability and M are $2.9 \times 10^{-18} \text{ m}^2$ and 2.2, respectively. Thus, despite the strong changes in fluid and current in this fracture network, the permeability and resistivity are still low, and this fracture network is below the percolation threshold.

The percolation threshold occurs at a fracture separation of -0.021 mm (mean aperture of 0.023 mm), where the first fully connected pathway for fluid flow forms through the network. For current, there is only one fully connected pathway, but several pathways that connect partially have now developed. There is also a small further decrease in both the current and fluid flow in the matrix. The percolation threshold is associated with a contact area of 60% and a fracture porosity of 1.1%. The permeability increases markedly here ($1.3 \times 10^{-15} \text{ m}^2$), while M shows another steady increase to 3.5.

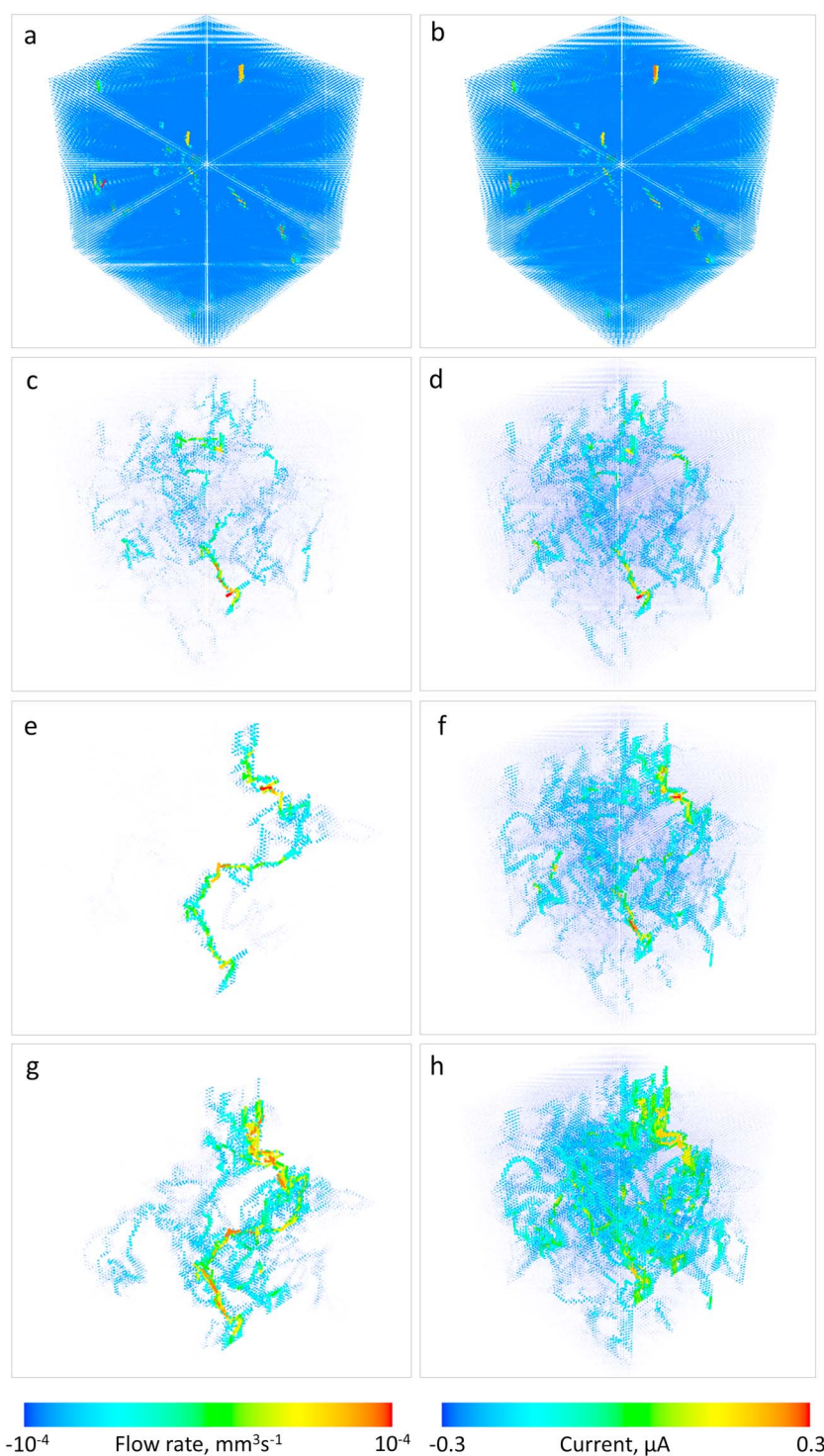


Figure 5. Fluid flow and current passing through a fracture network as the fractures are progressively opened. The network was built using an α value of 30. (a) Flow rate and (b) current for a fracture separation of -0.16 mm, (c) flow rate and (d) current for a fracture separation of -0.038 mm, (e) flow rate and (f) current for a fracture separation of -0.021 mm, and (g) flow rate and (h) current for a fracture separation of 0.009 mm. The resistivity and permeability of this network are shown in Figure 3.

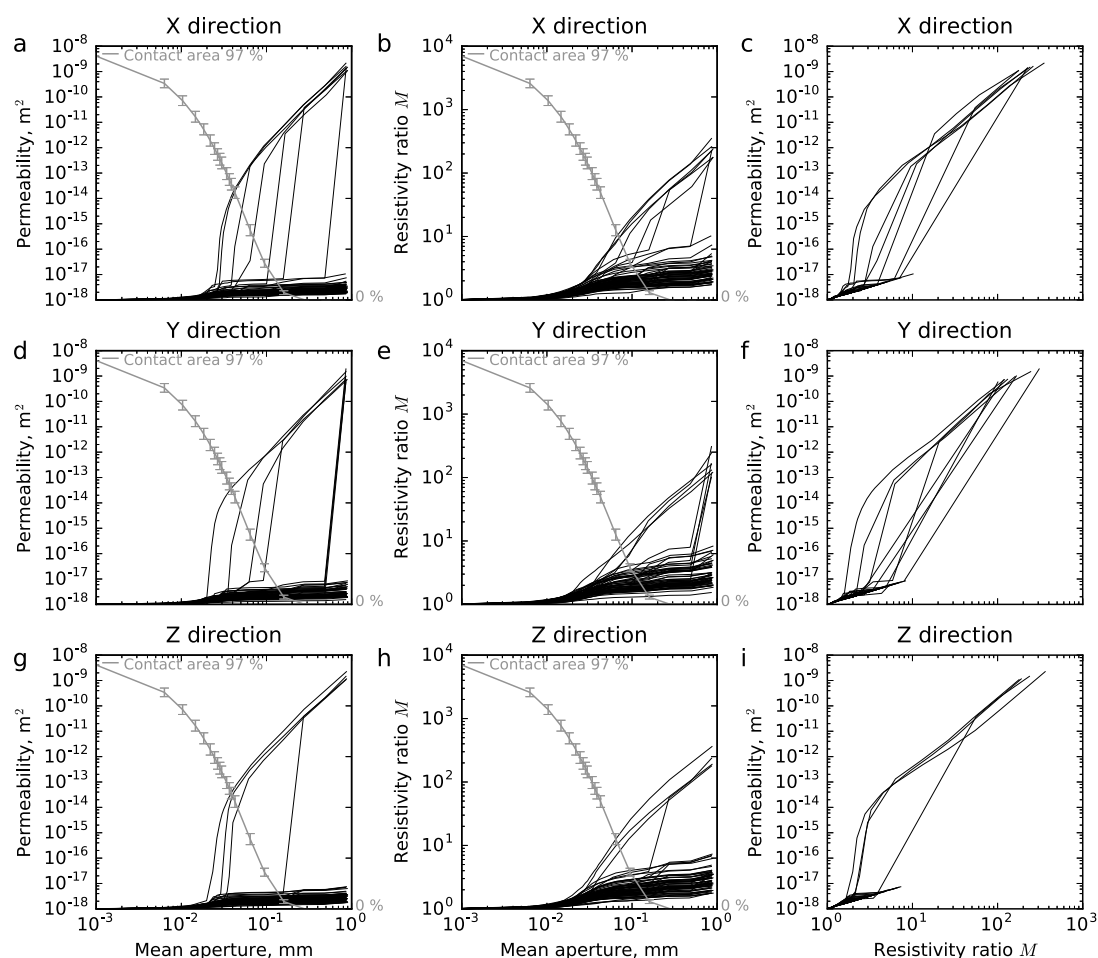


Figure 6. The evolution of permeability and resistivity in the x , y , and z directions (black lines) in a sparse fracture network with the density constant α equal to 3, as the fractures within that network are opened. (a, d, and g) Permeability and (b, e, and h) resistivity ratio M , in three directions as a function of arithmetic mean aperture. The mean and standard deviation contact area for all models is shown in grey. (c, f, and i) Permeability as a function of M .

By a fracture separation of 0.009 mm (mean aperture of 0.036 mm), multiple connecting fluid pathways have formed through the network, and the fracture porosity has increased to 1.8%. The contact area is now 46%, while the permeability and M values are $5.0 \times 10^{-14} \text{ m}^2$ and 8, respectively.

4.2. Sparse Fracture Networks

Figures 6 and 8 show resistivity and permeability as a function of aperture for α values of 3 and 0.3. The key difference in the results here is that for an α value of 3, only some of the networks demonstrate percolation behavior, and for an α value of 0.3, only one of them does. This makes intuitive sense: the parameter α controls the density of fractures, and as the density of fractures is decreased, the likelihood of having a fully connected fracture network also decreases. For an α value of 3, none of the fracture networks are fully connected (and thus able to demonstrate percolation behavior with respect to the fracture aperture) in all three directions; however, 16 of them, or 27%, have a percolation threshold in at least one direction. This implies that these rock volume models will have moderate to strong resistivity anisotropy (up to a factor of 160), and strong permeability anisotropy (up to a factor of 10^9), above the percolation threshold, since the rock is fully connected in some directions and not in others (Figure 7). However, there is no preferred orientation for which directions percolate and which do not, and so at larger scales this anisotropy may disappear.

For α equal to 3, there is a greater spread in the percolation threshold for both permeability and resistivity than for an α value of 30 (Figure 6). This results from the fact that many of the fracture networks for α equal to 3 are themselves close to a percolation threshold, i.e., even if all the fractures within the network are completely open, the networks themselves are on the verge between being connected and disconnected. Some of the

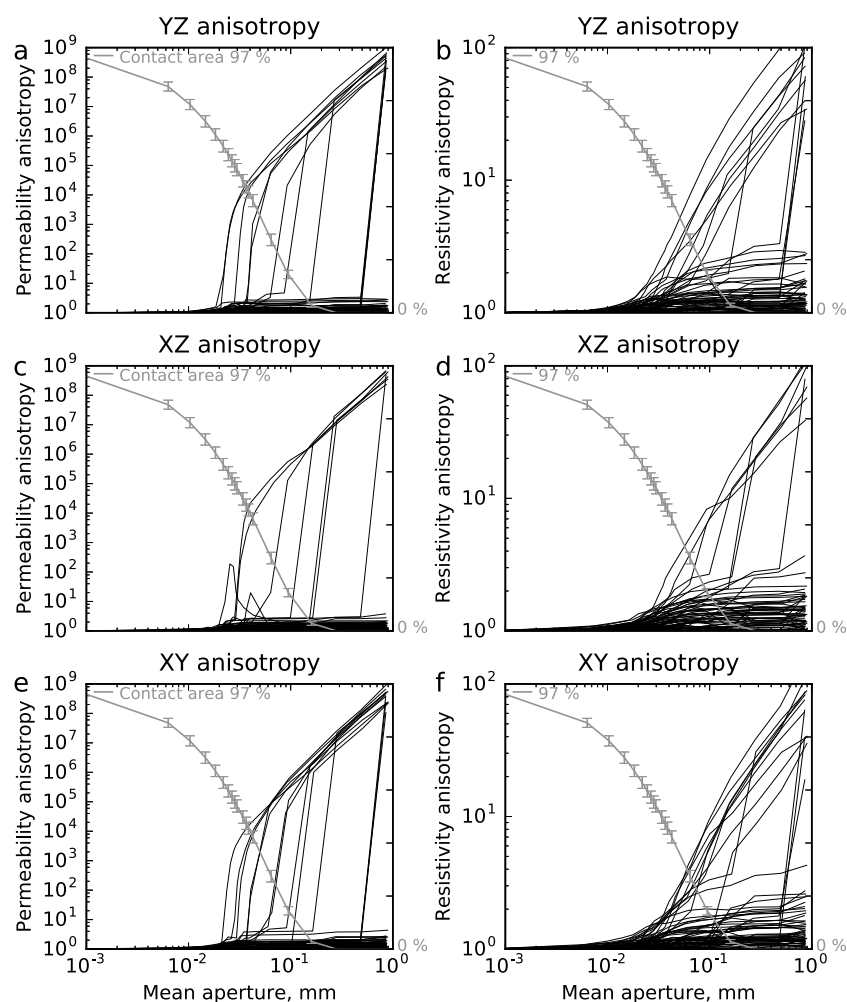


Figure 7. Anisotropy in permeability and resistivity as a function of arithmetic mean aperture for an α value of 3. Anisotropy shown as the maximum factor, i.e., xy anisotropy is the maximum of x/y and y/x resistivity or permeability. (a), c, and e) Permeability anisotropy. (b, d, and f) Resistivity anisotropy.

fracture network models may require that all the fractures within the network are fully connected, before the entire network will become connected. These will display a percolation threshold at a higher mean fracture aperture. Others may become connected more easily and will thus percolate at lower mean apertures.

5. Discussion

5.1. Summary of Results and Significance

Three-dimensional electrical conduction and fluid flow models of fluid-filled fracture networks have been developed to assess the effect that fractures have on the resistivity and permeability of a rock. In these models, the ratio of rock matrix to fluid resistivity, m , has been held constant at 10^4 . Similar to individual fractures, fracture networks also display percolation behavior.

The constant, α , which controls fracture density, is a first-order control on whether the fractured rock will be able to reach a percolation threshold. If α is low, then even when the individual fractures are fully connected, the resistivity and permeability will never reach values more than 1 order of magnitude different from the rock matrix values. On the other hand, if α is high, then the network can reach low resistivity and high permeability values, if the fractures themselves are sufficiently open.

The observation that fracture networks demonstrate a percolation threshold, below which the network is disconnected, has been noted in previous studies, most of which relate to fluid flow. A commonly applied method is Discrete Fracture Network (DFN) modeling, in which all the fluid flow is assumed to occur through

the fractures [e.g., Mourzenko *et al.*, 2004; Khamforoush *et al.*, 2008; Ahmed Elfeel *et al.*, 2012]. These studies have demonstrated that increasing the density of fractures in a network, which are controlled by the parameters α and a (section 2.3), increases both the probability of percolation in a fault network and its permeability.

Several authors have demonstrated the importance of incorporating variations in aperture within fractures in a network [e.g., Paluszny and Matthai, 2010; de Dreuzy *et al.*, 2012]. In both of these studies, permeability was overestimated when fractures were assigned an equivalent average aperture rather than variable aperture within the fracture plane. This difference was attributed to fluid pathways being blocked in some places in the variable aperture models.

The modeling presented here incorporates aperture variations but builds on these studies by also looking at the effect of incrementally separating the fracture planes. The present study shows that when the fracture density constant, α , is sufficiently high, the separation between individual fracture planes is a strong control on the resistivity and permeability of the fractured rock. Furthermore, even in an individual fracture network, the percolation threshold occurs at different fracture separation values for different orientations. Thus, near the percolation threshold, many of the models demonstrate anisotropy in both resistivity and permeability, even though the attributes of the fracture network (probability of a fracture in a given direction, size of fractures with different orientations) were assigned with equal probability for the xz , yz , and xy planes. Anisotropy is more prevalent at smaller values of α . For an α value of 3, around 27% of fracture networks reach a percolation threshold but only in one direction or two directions. This implies that above the percolation threshold, these will always have strong resistivity and permeability anisotropy. This effect may simply reflect the limited size of the fracture network and will likely disappear on larger fracture networks; however, it is interesting to note how easily anisotropic resistivity and permeability can develop in fractured rocks. If the fracture separation was allowed to vary with orientation, for example, then anisotropic properties may persist to larger volumes.

This work also builds on many previous studies by considering the permeability and resistivity of the rock matrix in the modeling. In many previous DFN and resistor network models [e.g., Long and Witherspoon, 1985; Berkowitz, 1995; Bahr, 1997; Mourzenko *et al.*, 2004; Khamforoush *et al.*, 2008], it has been assumed that the rock matrix is impermeable and infinitely resistive, so when the fracture network is not fully connected the permeability (and resistivity) is zero. However, several authors [e.g., Paluszny and Matthai, 2010; Ahmed Elfeel *et al.*, 2012; Kirkby *et al.*, 2016] have noted that significant fluid and current can pass through the rock matrix. Figure 5 confirms this, demonstrating that even at low rock matrix permeabilities and conductivities, and particularly below the percolation threshold, the rock matrix has an important role to play. Connecting pathways begin to form even below the percolation threshold, and these have an effect on both the permeability and resistivity of that network, even before it is fully connected. In the present study, only a low matrix permeability was considered; however, in previous work [e.g., Paluszny and Matthai, 2010; Kirkby *et al.*, 2016] it was shown that increasing the matrix permeability reduces the difference between the permeability and resistivity of the fracture and that of the matrix and so the percolation threshold is less well defined.

The changes in the resistivity as the individual fractures within fracture networks are opened will also depend on the ratio of rock to fluid resistivity, m (Figure 8). The control of this ratio on the overall rock resistivity was not analyzed here. Kirkby *et al.* [2016] showed that the decrease in resistivity at the percolation threshold becomes sharper with increasing m . This is because as m increases, the relative contribution of the matrix to the overall conduction decreases as the current becomes more focused in the fractures. This effect is less in the fluid flow, partly because of the cubic dependence of flow rate on aperture and partly because the difference in permeability between the rock and fractures tends to be higher than that for electrical resistivity. In any case, a similar effect is likely to be present in the fracture network as a whole, as in the fracture itself.

Fracture surface displacement was also not included in the models presented here. The displacement (offset) on a fracture depends on the style of fracturing and geological setting, but generally increases with fracture size, and is normally between 0.1 and 10% of size for fractures less than 1 m in length [Kim and Sanderson, 2005]. Thus, for the fractures considered here, offsets of up to 1.5 mm might be expected. In the fracture models considered by Kirkby *et al.* [2016], increasing the offset from 0 to 10 mm increased the maximum width of the fracture, which results in both the resistivity and permeability of fractures increasing by a greater amount at the percolation threshold. A similar effect may occur if offset is added to fractures within a network.

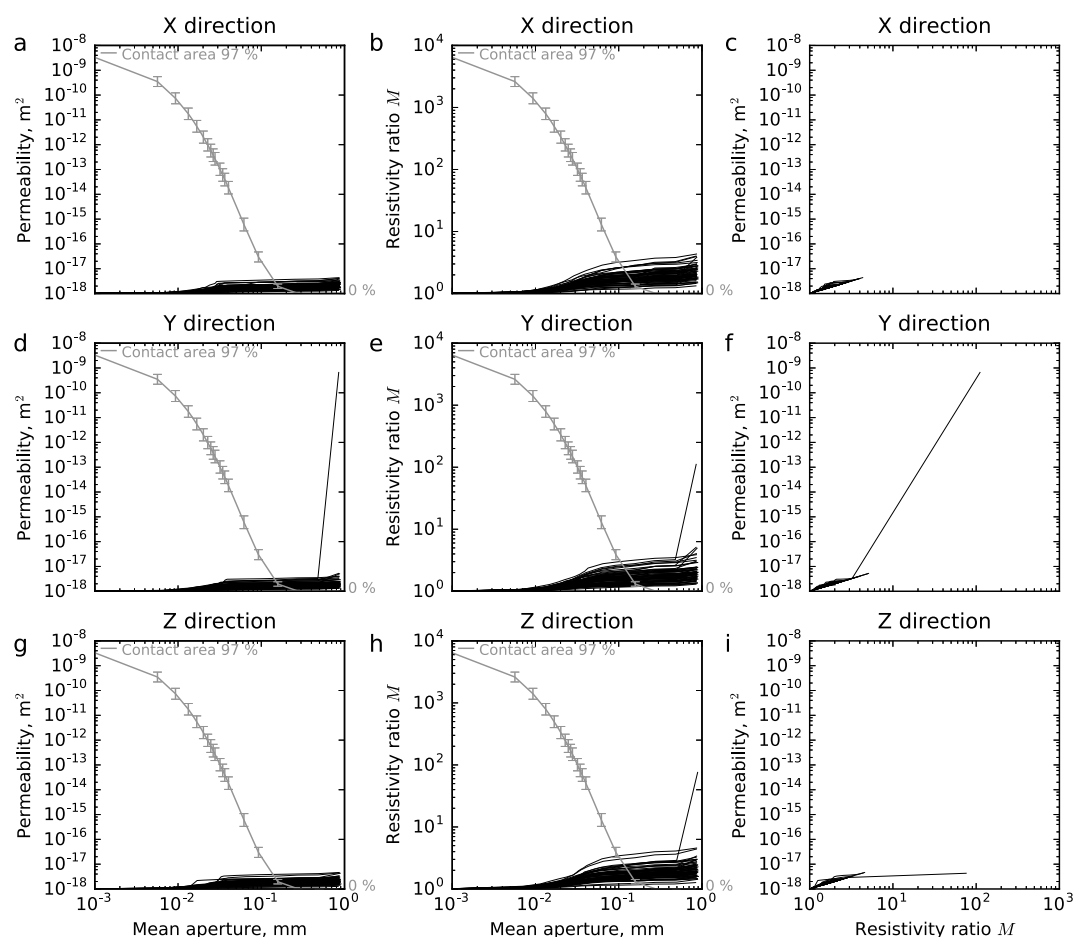


Figure 8. The evolution of permeability and resistivity in the x , y , and z directions (black lines) in a sparse fracture network with the density constant α equal to 0.3, as the fractures within that network are opened. (a, d, and g) Permeability and (b, e, and h) resistivity ratio M , in three directions as a function of arithmetic mean aperture. The mean and standard deviation contact area for all models is shown in grey. (c, f, and i) Permeability as a function of M .

5.2. Limitations and Real-World Applicability

There are several aspects of the approach used in this study that may limit the applicability of the results to a real-world situation. These would need to be addressed before the results can be applied more widely. Broadly, these limitations fall into three categories: (1) intrinsic limitations of the resistor network approach, (2) limitations in the fracture representations used here, and (3) model size limitations resulting from the model discretization.

5.2.1. Limitations of Resistor Network Approach

As discussed in section 2.1, the resistor network approach applied here is a simplification of the electric and hydraulic processes occurring in a fractured rock volume. The approach assumes laminar fluid flow and neglects both the frequency dependence and surface conduction component of resistivity. Thus, the results are most relevant to rocks without significant clay content and high fluid salinities, although they should be applicable to most natural groundwater flow rates (section 2.1).

5.2.2. Fracture Representation

The fracture surface model used in this study was developed based on isotropic fractures in crystalline rock [Brown, 1989; Matsuki *et al.*, 2006]. It does not incorporate many structures that commonly develop on fracture surfaces in response to fracture growth, such as hackles and rib marks [e.g., Pollard and Aydin, 1988]. The characteristics of such features depend on a number of factors such as the mechanical properties and geometry of the rock in which the fracture developed and the local stress conditions [e.g., Pollard and Aydin, 1988; Olson *et al.*, 2009]. The effects of such features on the permeability and resistivity of fractures could be addressed through a systematic study or through analysis of a specific field site, which could include mechanical modeling of fracture development.

5.2.3. Size Limitations

A key limitation of the grid-based approach used in this study arises from the computational demands from the large number of cells required to perform the calculation. In this work, we selected a small cell size of 0.25 mm in order to capture small-scale variations in fracture surface roughness and were therefore limited to modeling 1.5 cm cubes. A 1.5 cm cube may be representative of small core samples; however, it is not sufficient to adequately capture the properties of larger-scale fracture networks. In order to provide models that are more relevant to resistivities that are measured in well logs and by surface electrical geophysical methods, it would be necessary to scale up to larger rock volumes.

In order to scale up, there are a few ways in which the approach could be improved. These include use of a different solver and improvements to the meshing to reduce the number of grid cells in the computation.

An embedded approach similar to that used by Bahr [1997] could also be implemented. In this approach, each resistor in a larger network consists of a new, smaller network, which is analyzed separately. This could be applied to the type of modeling presented here by first generating a fracture network at the desired scale and dividing that network into smaller cubes. The resistivity and permeability of these individual cubes could then be computed and used to fill the larger rock volume. This embedding process could be repeated multiple times, such that the faults are analyzed to a smaller and smaller level of detail. In this way, both the small-scale fracture surface roughness and the geometric properties of larger fault networks would be captured.

6. Conclusions

Modeling of 3-D fracture network models has shown that the percolation behavior that was found in individual fractures is also present in fracture networks, as the fractures within that network are progressively opened. The ability of a network to percolate depends on characteristics of the fracture network itself, notably the density constant, α , as well as the properties of the individual fractures within the network, particularly the separation between the fracture planes.

In the fracture networks presented here, where the rock to fluid resistivity ratio is equal to 10^4 , resistivity enhancements up to a factor of 5000 and permeability enhancements up to 10 orders of magnitude were observed. However, as with the individual fracture planes, permeability enhancement greater than 1 order of magnitude and resistivity enhancement greater than a factor of 2–3 is only possible if the fracture network is above its percolation threshold. This requires both the network itself and the fractures within it to be connected.

Resistivity anisotropy is common in fracture networks near their percolation threshold. In fracture networks with α equal to 3, more than half of them show anisotropy even when all the fractures within the network are completely open, because the fracture network itself is near a percolation threshold. Likewise, many of the dense fracture networks, with α equal to 30, display anisotropy when the fractures within that network are close to their percolation threshold. This effect may disappear on larger fracture networks; however, it demonstrates how easily anisotropic properties can appear. Future work could look at how anisotropy in fracture networks is controlled by variations in the fracture separation with orientation.

Acknowledgments

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